

Himal Shrestha

PHD, BIOINFORMATICIAN, MDU-PHL

The University of Melbourne, Australia

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A Bioinformatician in MDU-PHL at the University of Melbourne with a PhD in Bioinformatics, passionate about using data to predict, prevent, and eliminate infectious diseases. My research and interest lies in the integration of genomic, environmental, and epidemiological data with a variety of modelling and analytical approaches to study the transmission of infectious diseases.

Education

PhD in Bioinformatics

LA TROBE UNIVERSITY

Melbourne, Australia

2019-2023

- **Thesis:** New tools for onchocerciasis elimination in Africa: the role of spatially explicit models in informing decisions.

Master of Biotechnology and Bioinformatics

LA TROBE UNIVERSITY

Melbourne, Australia

2017-2019

Bachelor of Science in Human Biology

KATHMANDU UNIVERSITY

Dhulikhel, Nepal

2012-2016

Skills

COMPUTATIONAL SKILLS

- Programming languages: Python, R, MATLAB, Linux Shell, Bash, Nextflow, Snakemake, Julia, javascript and C; Markup and version control: LaTeX, Markdown, YAML, Git, GitHub, Bitbucket, VSCode

Professional experience

Bioinformatician

MDU-PHL, THE UNIVERSITY OF MELBOURNE

Melbourne, Australia

Oct, 2023-Ongoing

- Developing and applying computational tools to analyse genomic data of bacterial pathogens

Post-doctoral researcher (Bioinformatics/infectious disease modelling)

LA TROBE UNIVERSITY

Melbourne, Australia

Apr, 2023-Oct, 2023

- Formulating transmission models utilising genomic, environmental and epidemiological data and supervise students.

Publications

SELECTED PUBLICATIONS

- **Shrestha H**, McCulloch K, Chisholm RH, Armoo SK, et al. Synthesizing environmental, epidemiological and vector and parasite genetic data to assist decision making for disease elimination. Cotton J, editor. Molecular Ecology. 2024 Jun 33;11.
- Panikulam EJ, Seenarain A, **Shrestha H**, Wan Chin JL, et al. A case report of *Herbaspirillum* infection in rural Australia. ASM Case Reports. 2025 Mar 1;3.
- **Shrestha H**, McCulloch K, Hedtke SM, Grant WN. Geospatial modeling of pre-intervention nodule prevalence of *Onchocerca volvulus* in Ethiopia as an aid to onchocerciasis elimination. Cotton J, editor. PLoS Negl Trop Dis. 2022 Jul 18;16(7):e0010620.
- More publications can be found at [Google Scholar](#)

Research grants

Co-investigator/Sub-awardee

UNICEF/UNDP/WORLD BANK/WHO SPECIAL PROGRAMME FOR RESEARCH AND TRAINING IN TROPICAL DISEASES (TDR/WHO)

(US\$108,000)

2021-2022

- Population genetic tools for onchocerciasis control programmes to determine transmission zones.

Co-investigator/Sub-awardee

CENTER FOR PATHOGEN GENOMICS (CPG) AT THE UNIVERSITY OF MELBOURNE (AU\$30,000)

2026-2027

- Genomic bioprospecting of antimicrobial, probiotic, and extremophilic bacteria from Nepal.